mPing k-mer screen: preliminary results

Generated 2026-10-23 from code at commit b584e93 (uncommitted changes present).

Copy numbers of rice transposable elements (TEs) estimated from k-mer depth in short reads, compared with published values. The algorithm is described in `docs/ALGORITHM.md`.

Scope. k=21; 12 strains (CX55 has no baseline and no copy numbers); 14 TEs in `RiceTE.fa`; no GC correction applied.

Summary

- For the 9 low-mPing strains, the k-mer mPing estimate and the Chen et al. 2019 read-depth estimate have Pearson r = 0.969 and Spearman 0.979. The k-mer value is a median 1.12 times the paper's (range 0.94 to 1.48).
- Nipponbare (NB) gives 52.4 mPing copies. The paper lists 52 for its Nipponbare entry.
- HEG4 gives 485. The paper reports 231 to 503 for its four bursting accessions and no value for HEG4 alone.

Figure 1. mPing copies against the paper

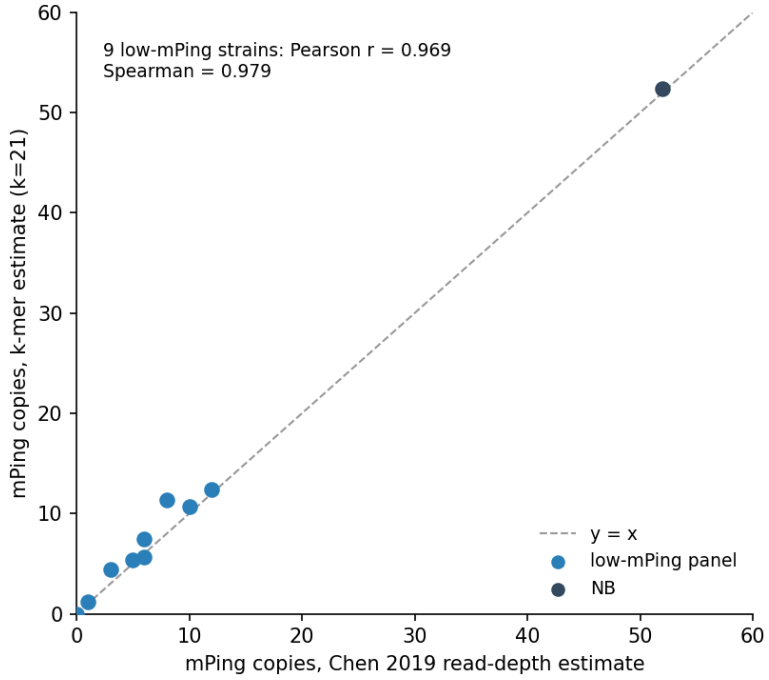


Figure 1: mPing k-mer estimate against the Chen 2019 estimate

With 9 points, the correlation is a small-sample result.

Figure 2. mPing copies by strain

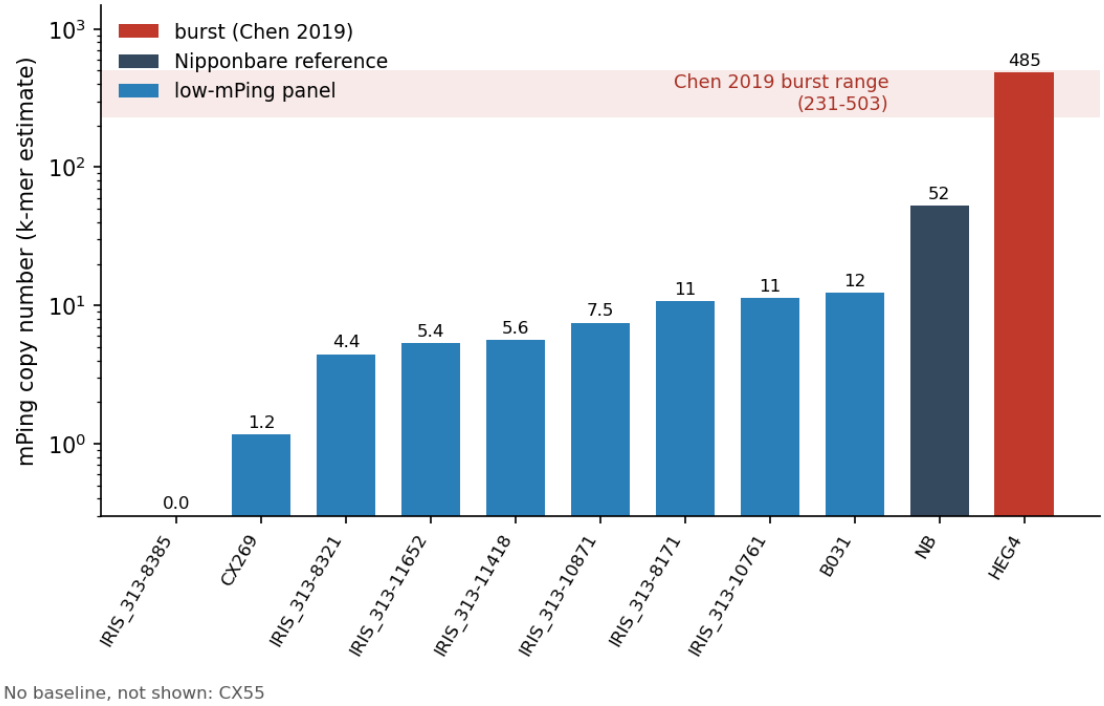


Figure 2: mPing copy number by strain

The low-mPing panel spans 0 to 12 copies, NB has 52 and HEG4 has 485. Fold change against NB understates how low the panel is, because NB carries 52 copies. The HEG4 value has no independent check.

strain	group	single_copy_depth	paper_mPing	kmer_mPing_k21	ratio
HEG4	burst	133		484.6	
NB	reference	57	52	52.4	1.01
B031	low_panel	8	12	12.4	1.03
IRIS_313-10761	low_panel	6	8	11.3	1.42
IRIS_313-8171	low_panel	7	10	10.7	1.07
IRIS_313-10871	low_panel	12	6	7.5	1.25
IRIS_313-11418	low_panel	8	6	5.6	0.94
IRIS_313-11652	low_panel	17	5	5.4	1.07
IRIS_313-8321	low_panel	9	3	4.4	1.48
CX269	low_panel	12	1	1.2	1.17
IRIS_313-8385	low_panel	5	0	0.0	
CX55	low_panel		10	n/a (no baseline)	

Figure 3. Position profile along mPing

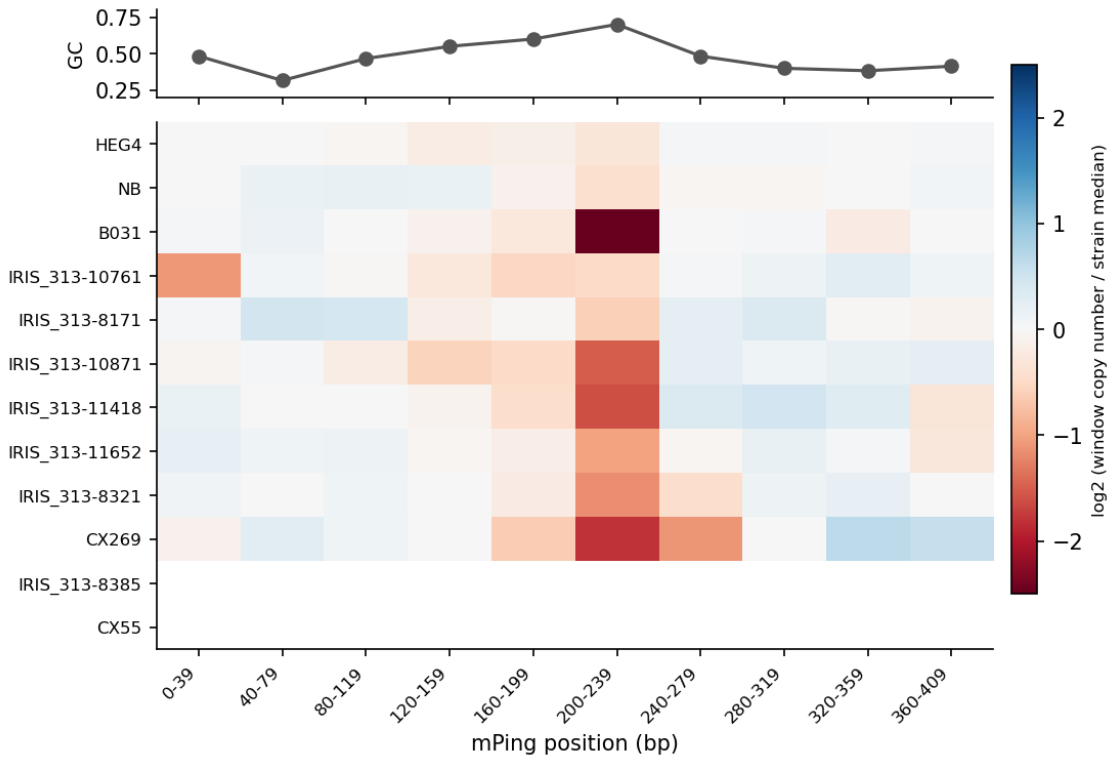


Figure 3: mPing window copy number along the element

Windows are 40 consecutive unique k-mers. The window at positions 200 to 239 is the highest-GC stretch of mPing (GC 0.70, against 0.47 for the whole element). Several strains show a dip in this window. The lowest window is below 0.4 of the strain median in B031 (0.17), IRIS_313-11418 (0.33), CX269 (0.29), IRIS_313-10871 (0.35). A dip shared across strains at the highest-GC window suggests a sequencing bias. This has not been tested. The GC correction is implemented in `scripts/mitekmer.py` and has not yet been applied to these results. A strain with no mPing k-mers has an empty row.

Figure 4. All TEs in the library

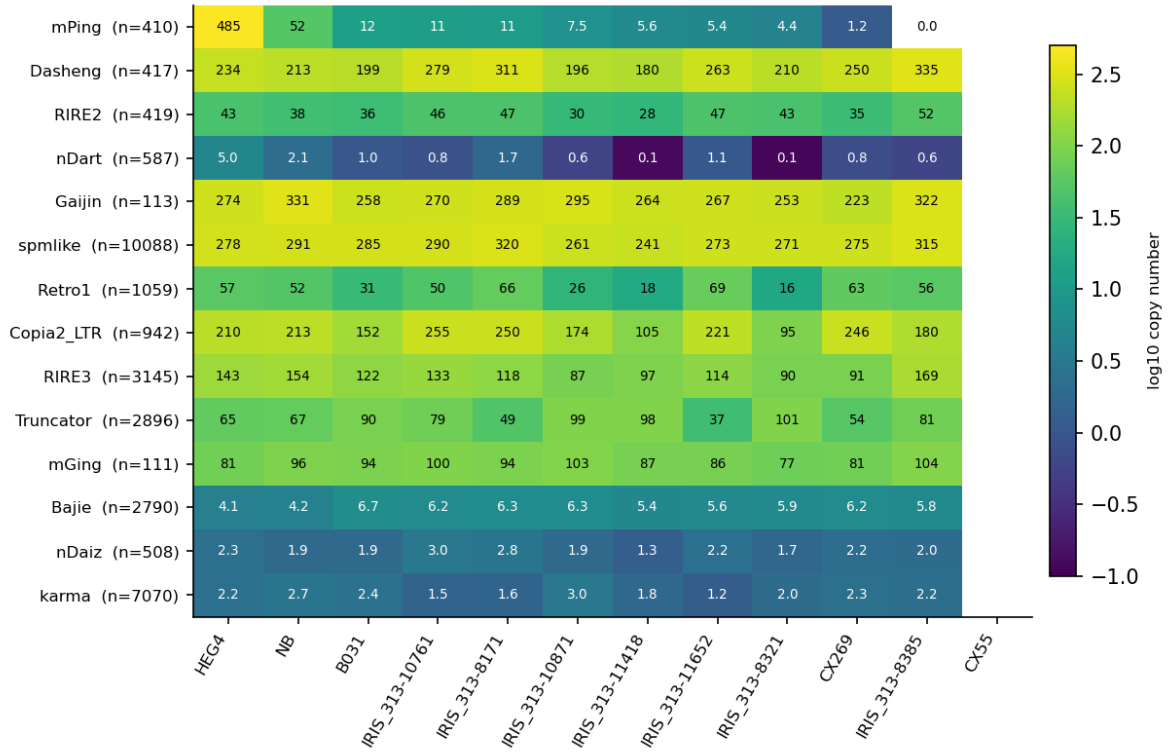


Figure 4: Copy number of every TE in every strain

For the TEs other than mPing and nDart, the largest-to-smallest ratio across strains ranges from 1.3 to 4.2. For mPing the ratio is about 415 (485 against 1.2, leaving out strains with 0 copies). nDart: 5.0 in HEG4 against 0.1 to 2.1 elsewhere. There are no published copy numbers for these TEs in the data used here, so the remaining spread cannot be assigned to biology or noise. `n` is the number of unique k-mers per TE.

Limits

- Each strain is scaled by its own single-copy depth, taken as the highest peak above the error valley of its k-mer histogram. Low-depth strains (single-copy depth 5 to 9) sit close to that valley, so their scaling is less certain.
- A k-mer must match exactly, so diverged copies are undercounted.
- A strain whose k-mer histogram has no clear single-copy peak gets no copy numbers instead of a guess (the peak must be at least 1.1 times the error valley, with the valley at count 200 or below).
- See `docs/ALGORITHM.md` for the full list.

Not yet included

- GC-corrected copy numbers.
- k=31 results for the same strains.
- Landraces A119, A123, A157 and CX140 (the paper's Nipponbare entry, streamed from ENA).
- 100 additional lines from the 3,000-genome set.